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Transjakarta Passenger Estimates for Early July 2024 using MRT Route Impact with Autoregressive Distributed Lag

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Abstract
Transjakarta is a transportation system initiated by the Provincial Government of DKI Jakarta since 2014. Understanding the number of passengers who will use Transjakarta is a strategic measure, especially following route adjustments due to the construction of MRT Jakarta Phase 2A CP 202 during the period from March to July 2023. This study employs the Autoregressive Distributed Lag (ARDL) method to forecast the number of Transjakarta passengers at the beginning of July 2024. The data used consists of daily Transjakarta passenger counts from January 2023 to June 2024. The data is divided into 87% for training and 13% for testing. The optimal model formed has lags of p = 1 and q = 3. The model satisfies all assumption tests, including normality, homogeneity, autocorrelation, and zero mean error expectations. The Mean Absolute Percentage Error (MAPE) on the training and testing data are 21.69% and 11.19%, respectively. These values indicate that the forecasted values for both training and testing data are close to the actual data and do not exhibit overfitting. The forecast for the next 14 days reveals a dynamic pattern with sharp changes followed by gradual stabilization.

Keyword: DKI Jakarta, forecasting, MAPE, route MRT, Transjakarta

Abstrak
Transjakarta adalah sarana transportasi yang diinisiasi oleh pemerintah Provinsi DKI Jakarta sejak tahun 2014. Mengetahui jumlah penumpang yang akan menggunakan Transjakarta adalah hal strategis yang dapat dilakukan, terutama setelah terjadinya penyesuaian rute akibat konstruksi MRT Jakarta Fase 2A CP 202 pada periode Maret hingga Juli 2023. Penelitian ini menggunakan metode *Autoregressive Distributed Lag (*ARDL) dengan tujuan melakukan peramalan jumlah penumpang Transjakarta pada awal bulan Juli tahun 2024. Data yang digunakan adalah data harian jumlah penumpang Transjakarta dari bulan Januari 2023 hingga Juni 2024. Data dibagi dengan 87% data *training* dan 13% data *testing*. Model yang dibentuk memiliki lag optimal yaitu p = 1 dan q = 3. Model tersebut memenuhi seluruh uji asumsi, yaitu normalitas, homogenitas, autokorelasi, dan nilai harapan galat sama dengan nol. *Mean Absolute Percentage Error* (MAPE) pada data *training* dan *testing* adalah sebesar 21.69% dan 11.19%. Nilai tersebut menunjukkan bahwa peramalan dari data *training* dan *testing* mendekati nilai dari data aktual dan tidak mengalami *overfitting*. Peramalan dilakukan pada 14 hari ke depan yang menghasilkan pola dinamis dengan perubahan tajam dan stabilisasi bertahap.

Kata Kunci: DKI Jakarta, peramalan, MAPE, rute MRT, Transjakarta

INTRODUCTION

Transportation facilities are very important to ease people's lives and develop businesses in order to achieve access to basic services both in rural and urban areas (Cakrawijaya et al., 2014). In its development, public transportation serves as an alternative to fulfill community needs, especially in large cities like DKI Jakarta Province. DKI Jakarta is the province with the highest population density in Indonesia with 15,978 people per square kilometer (BPS, 2021).The government initiated a solution by creating the Transjakarta Busway. The implementation of the Transjakarta service is based on one of the legal products, namely *undang - undang* No. 14 of 1992 concerning *pasal* 12 of road traffic (Siti et al., 2020). In its operations, Transjakarta has served approximately 13 corridors with a total route length of 215,200 km in DKI Jakarta. Of course, this is a concrete from of the government's efforts to address issues and meet the needs of the community (Kevin & Mariah, 2022).

Route adjustments made by PT Transjakarta during the period of March to July 2023 due to the construction of Mass Rapid Transit (MRT) Jakarta Phase 2A CP 202 (PT MRT Jakarta (Perseroda), 2022), have a significant impact on service operations. One of the most striking impacts is a drastic decrease in the number of passengers every day during the adjustment period. This phenomenon is an important concern because it can affect the level of public transportation use in DKI Jakarta and the effectiveness of transportation policies implemented during the construction period. Conducting a time series forecasting analysis for the number of Transjakarta passengers after this route adjustment is an interesting and relevant step. With this analysis, predictions can be made about the number of passengers in the future. This analysis can provide valuable insights for PT Transjakarta and local governments in designing strategies to increase the number of passengers, including better mitigation measures to reduce negative impacts when the project construction period occurs again.

Time series analysis is one of the approaches often used in research to understand and predict data patterns based on time sequence. The goal of time series data analysis is to understand patterns that occur in data measured sequentially over time (MY et al., 2024). Time series data, commonly used across fields like economics, business, engineering, natural sciences, and social sciences, is characterized by interdependent observations over time, making the analysis of their dependency patterns particularly compelling (Ridwan et al., 2023). One example of the application of the Long Short-Term Memory (LSTM) method is a study by Davi (2018) entitled "Forecasting the Number of DKI Jakarta Busway Passengers Based on Smart Card Data Using Long Short-Term Memory" and another study by (Davi & Winarko, 2023)entitled "Design and Development of a Passenger Forecasting Application Using Long Short-Term Memory (LSTM)." The first study utilized smart card data to predict the number of busway passengers in DKI Jakarta, while the second focused on developing an application designed to forecast passenger

numbers using the LSTM method. The LSTM method was chosen because the data used is univariate, which only involves one main variable (Prasetyo et al., 2024).

The route adjustment policy that affects the fluctuation of the number of passengers is a challenge in itself in this study, so relying only on univariate data is considered less representative. To improve the accuracy of predictions, another approach is needed that is able to include additional relevant variables. This study uses the Autoregressive Distributed Lag (ARDL) method, because it will be able to analyze the dynamic relationship between the main variables and additional variables more comprehensively.

According to Gujarati & Porter (2012), ARDL is a method aimed at analyzing the influence of independent variables on dependent variables over time, including the impact of past dependent variables on the present. This study employs the independent variable of route adjustment and the dependent variable of the number of Transjakarta passengers. Based on this background, the purpose of this study is to implement the ARDL method in forecasting the number of Transjakarta passengers at the beginning of July 2024.

METHOD

The time series data used in this research is the number of the Transjakarta passengers with a daily period starting from January 1, 2023 to June 29, 2024. The number of Transjakarta passengers data is secondary data with 546 periods obtained from the https://data.go.id (*Portal Satu Data Indonesia*). This study uses the dependent variable, which is the daily number of Transjakarta passengers and the independent variable, which is the adjustment of the Transjakarta routes represented by binary numbers 0 (no adjustment of Transjakarta routes) and 1 (adjustment of Transjakarta routes).

Data analysis in this study was conducted using R-Studio software through several stages. The first stage is data exploration to identify patterns and characteristics. Then, the data is divided based on pattern similarity to facilitate analysis. The determination of the optimum lag is carried out using the ARDL method based on the Akaike Information Criterion (AIC) (Nkoro & Uko, 2016). The ARDL method allows the dependent variable to be influenced by the independent variable at a certain time as well as the value of the dependent variable itself at a previous time (Gujarati, 2004). ARDL modeling is formulated as follows:

$$Y_t = \alpha_0 + \alpha_1 + \alpha_2 Y_{t-1} + v_t \quad (1)$$

with Y_t represents the dependent variable at time t, a_0 is the constant, a_1 is the coefficient for the independent variable, a_2 is the coefficient for the lagged dependent variable Y_{t-1}, and v_t is the error term at time t. The optimum lag is determined by examining the lag combinations of the dependent and independent variables to find the lag values p (the limited lag length for the dependent variable) and q (the order of the autoregressive process for the independent variable) that yield the lowest AIC value. The AIC formula used is formulated as follows:

$$AIC = \ln\left(\frac{\Sigma_{t=1}^{T} e_t^2}{T}\right) + \frac{2p}{T} \quad (2)$$

with $\sum_{t=1}^{T} e_t^2$ as the sum of squared residuals, T as the number of data points, and p as the number of parameters in the model (Montgomery et al., 2015).

Determination of the optimum lag is followed by the analysis of the Cross-Correlation Function (CCF) plot to identify the correlation between variables(Mahmoudi & Zarei, 2023). The CCF plot helps determine the significant time lag with a p-value < 0.05 (Chadsuthi et al., 2012). If the CCF plot is not informative enough, a manual iteration is performed by gradually adding lags until the lag is no longer significant, starting with the lag of the independent variable (X_t) and followed by the lag of the dependent variable (Y_t). Based on the significant lags, the ARDL model is then formed.

Diagnostic tests are conducted to ensure that the statistical assumptions of the ARDL model are met. The Durbin-Watson test checks for independence (no autocorrelation) (Toor & Islam 2019). The Box-Ljung (Dare et al., 2022) test checks for homoscedasticity of residuals. The Kolmogorov-Smirnov (Huang et al., 2007) test checks for normality. The t-test (Sari et al., 2017) checks if the expected error value is equal to zero. If the p-value > 0.05 in all those diagnostic tests, it can be concluded that the assumptions in the ARDL model have been met.

Forecasting is performed on the training data to obtain forecast values. This forecasting process uses the multistep forecast method, which is a prediction methodology aimed at estimating a series of values on a future horizon with h > 1, where h represents the number of periods to be predicted (Bai et al., 2024). After that, the forecasting continues with forecasting on the testing data.

Model validation is conducted by comparing the forecasting results with actual data. The model's accuracy is measured using the Mean Absolute Percentage Error (MAPE), which calculates the percentage deviation between actual data and forecasting results (Krisma et al., 2019). A smaller MAPE value indicates a more accurate forecasting model that closely approximates the actual value (Mulyani et al., 2022). The MAPE calculation follows the formula provided by Poon (2005):

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{X_t - \hat{X}_t}{X_t}\right| \times 100\% \quad (3)$$

with X_t as the actual value at time t and $\hat{X}_t$ as the forecast value at time t. As a final step, forecasting is applied for the beginning of July 2024 to predict the number of passengers in the future.

RESULTS AND DISCUSSION

The initial step in this research is to conduct data exploration using a time series plot. The data used is the daily number of Transjakarta passengers from January 1, 2023

to June 29, 2024. The visualization of the data exploration results is shown in Figure 1.

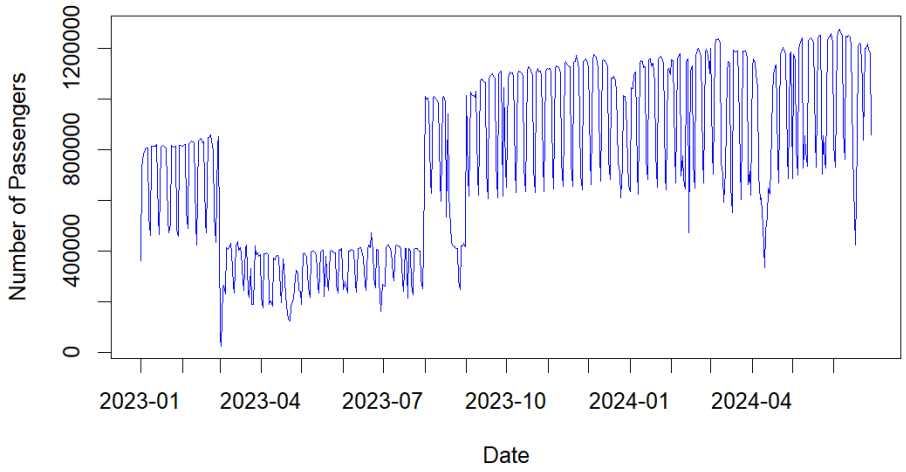


Figure 1. Time series plot of Transjakarta passenger numbers

Figure 1 above reveals significant patterns that offer insights into the usage of the Transjakarta service. In general, it appears that the number of Transjakarta passengers tends to increase over time, indicating growth in usage. During the period from March to July 2023, there was a significant decrease in the number of passengers. This is the impact of route adjustments due to the construction of MRT Jakarta Phase 2A CP 202, which likely caused temporary disruptions. After the July 2023 period, when the route adjustments were made, the number of Transjakarta passengers increased again, indicating that the public had adapted to the new route and stabilized after the route adjustments.

The data used in this study consists of 546 observations. This data is then divided into two parts, namely training data and testing data. The training data in this study amounts to 87% or approximately 477 observations, and the test data amounts to 13% or approximately 69 observations. The training data serves to standardize the model training process. Meanwhile, the testing data is used to verify the results of the model training (Andrianto 2018).

The determination of the optimal lag for the ARDL model is carried out using an iterative approach. Various combinations of lag p and q were tested to obtain the lag combination that produces the smallest AIC value as the best model criterion (Putri & Helma, 2024). The AIC value is calculated using Equation (2), which takes into account the sum of squared residuals, the number of parameters, and the number of data points. From the iteration results, the optimal lag combination was obtained, namely $p = 14$ and $q = 5$. Although this combination yields the smallest AIC value, the model does not meet the assumption of homoscedasticity. This indicates that the model has non-constant residual variance, so the lag selection needs to be reconsidered.

The CCF plot serves as an alternative to identify the relationships between variables at various lags. This analysis aims to find significant lags that influence the dependent variable. The results of the CCF plot can be seen in Figure 2.

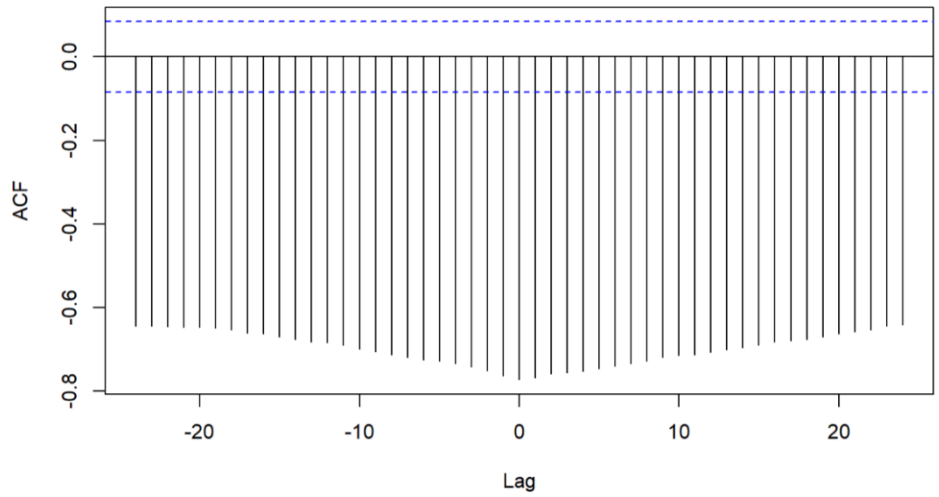


Figure 2. CCF Plot

Figure 2 shows that all lags exceed the confidence interval limits. This result indicates that all lags have a significant relationship with the dependent variable, making the CCF plot less effective in simplifying lag selection. Due to the two previous approaches not yielding optimal results, the determination of lag was carried out through manual iteration, which involved gradually introducing the lag and testing its significance. Lag is added to the model sequentially, starting from the lag of the independent variable to the lag of the dependent variable. Each addition is evaluated based on its significance. This process stops when the addition of lags is no longer significant, so the final model only includes significant lags. This approach results in an optimal lag combination with $p = 1$ and $q = 3$. Based on that combination, ARDL modeling was conducted to analyze the relationship between variables. The modeling results are presented in Table 1.

Table 1. Estimated parameters of the ARDL model

Variable	Estimated Parameters	p-value
Intercept	6.459×10^5	0.0000***
X_t	-6.818×10^5	0.0000***
X_{t-1}	4.326×10^5	0.0004***
Y_{t-1}	0.6944	0.0000***
Y_{t-2}	-0.2984	0.0000***
Y_{t-3}	0.1595	0.0003***

Based on the results of Table 1, it was found that the variables X_t, X_{t-1}, Y_{t-1}, Y_{t-2}, and Y_{t-3} has p-value < 0.05. This indicate that the variable X_t, X_{t-1}, Y_{t-1}, Y_{t-2}, and Y_{t-3} significantly affects the variable Y_t. In other words, route adjustments in the current period and the previous period, as well as the number of Transjakarta passengers in the previous period, two periods ago, and three periods ago, significantly impact the number of Transjakarta passengers in the current period. For the overall model, it is as follows:

$$Y_t = 6.459 \times 10^5 - 6.818 \times 10^5 \mathrm{X_t} + 4.326 \times 10^5 X_{t-1} + 0.6944 Y_{t-1} - 0.2984 Y_{t-2} + 0.1595 Y_{t-3} \quad (4)$$

The ARDL model with order $p = 1$ and $q = 3$ is further analyzed through a series of diagnostic tests. These diagnostic tests are conducted to ensure that the model meets basic assumptions, such as homoscedasticity, no autocorrelation, normality, and expected error value is equal to zero. The results of all the diagnostic tests are presented in Table 2.

Table 2. ARDL model diagnostic test

Assumptions	Test	p-value
Homoscedastisity	Box-Ljung Test	0.294
No Autocorrelation	Durbin-Watson Test	0.402
Normality	Kolmogorov-Smirnov Test	0.180
Expected error value is equal to zero	T-Test	1.000

The ARDL model meets the basic assumptions required to provide valid estimation results, as shown by the diagnostic tests in Table 2. The homogeneity of variance test using Box-Ljung yielded a p-value = 0.294 > α = 0.05 which means that the assumption of homogeneity of variance has been met. The autocorrelation test was also conducted using the Durbin-Watson test. This test resulted in a p-value = 0.402 > α = 0.05 indicating that the assumption of no autocorrelation between residuals is met. Next, a normality test was conducted using the Kolmogorov-Smirnov test, which resulted in p-value = 0.180 > α = 0.05 indicates that the normality assumption is met. Finally, a test of the expected error value being equal to 0 was conducted using the t-test. The result of the t-test showed a p-value = 1, thus confirming the assumption that the expected error value is equal to 0.

Testing data forecasting is conducted through iteration using multistep forecasting. The iteration proceeds by gradually updating the training data by adding data from the testing data up to a certain point in time. After the data has been updated, a one-step forecast is made, predicting the next value based on the current data. The results of the iteration contain all predictions for each time point in the testing data presented in Figure 3.

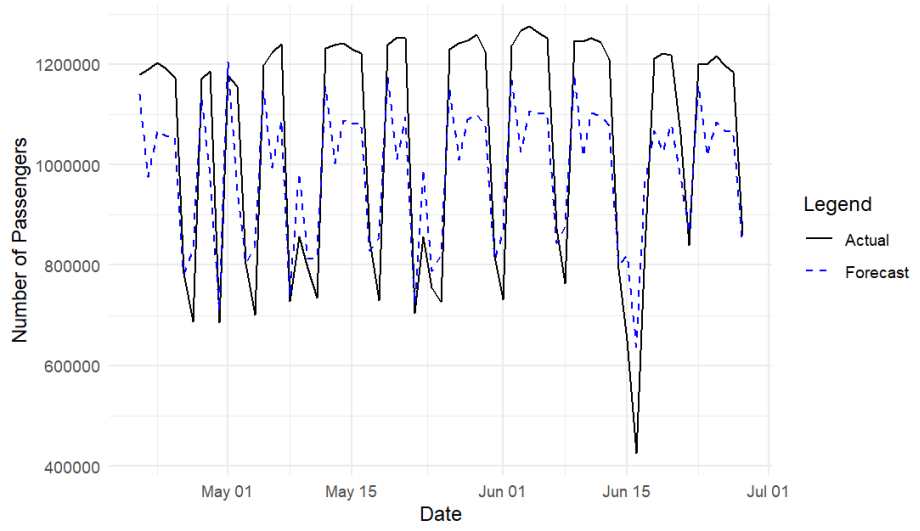


Figure 3. Plot testing data of forecast and actual results

Figure 3 shows the comparison between the actual data (black line) and the forecast results (blue dashed line) for the testing data over a certain period. From the plot, it can be seen that the forecast results generally follow the pattern of the actual data, including capturing weekly fluctuations and seasonal trends. However, at certain points, there are differences between the forecasted and actual values, especially during sudden changes that may be challenging for the model to accurately capture. This is a common occurrence in predictive modeling, as noted by(Amanu et al., 2024), who explain that such discrepancies can arise from various factors, such as abrupt changes in the data that are difficult to predict or limitations in the model's ability to represent all the dynamics present in the data.

Nonetheless, the relatively good alignment between the forecasted and actual lines demonstrates the model's capability to effectively capture the overall pattern. Model validation is conducted on the forecast data against the actual data by measuring accuracy using the MAPE value for both training and testing data. This MAPE value is calculated using the formula provided in Equation 3, and the results are presented in Table 3.

Table 3. MAPE values

MAPE	Value
Training	21.693%
Testing	11.191%

Table 3 shows that the MAPE values for the training and testing data fall into different accuracy categories. The MAPE value for the training data is 21%, which can be categorized as fair (adequate), while the MAPE value for the testing data is 11%, categorized as good (Sihombing et al., 2024). The difference in MAPE values between the training and testing data also not significant, which shows that the forecasting results for the training and testing data do not experience overfitting. Overfitting is a condition where the model can predict the training data very well but is unable to make good predictions for new, similar data (Tetko et al., 2024).

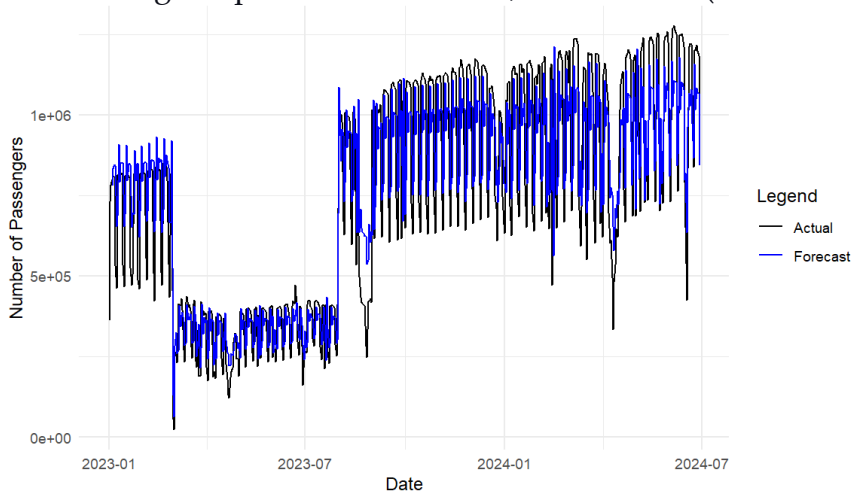


Figure 4. Plot the forecasted data and the actual data

Figure 4 shows the comparison between actual data (black line) and forecasted results (blue line) over the observed period. The forecasted values are very close to the actual data in both the training and testing periods. This indicates that the model effectively captures underlying patterns and seasonal fluctuations. The consistent alignment also confirms that the model generalizes well and does not experience overfitting. Overall, this figure demonstrates the model's ability to produce accurate forecasts.

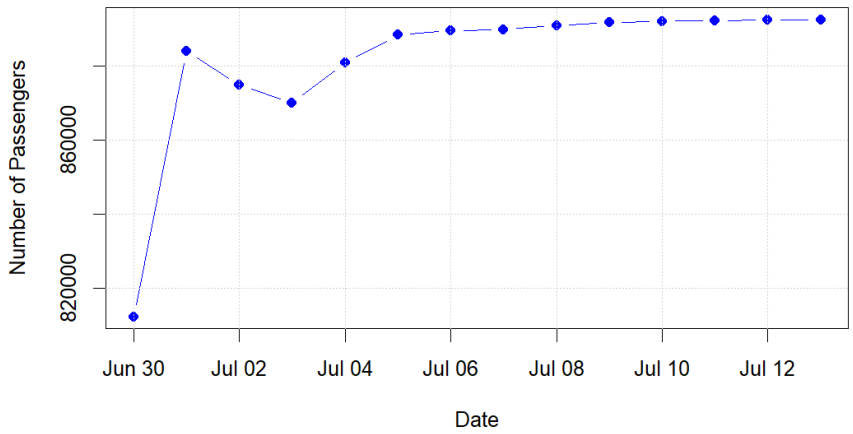


Figure 5. Plot forecasting for the next 14 days.

Figure 5 shows a sharp increase in the number of Transjakarta passengers from day 1 to day 2. From day 3 to day 5, fluctuations occurred where the forecasted values decreased and then rose again. After day 5 until day 14, the forecasted values tended to show a slight increase and exhibited a more stable pattern. This pattern suggests a dynamic trend in passenger numbers during the forecast period, reflecting both sharp changes and gradual stabilization. To further analyze the model's performance, Figure 6 presents a plot combining the 14-day forecasted values with the actual observed values. This visualization provides a clear comparison, allowing for an evaluation of forecast accuracy and highlighting any discrepancies between the predicted and actual passenger numbers.

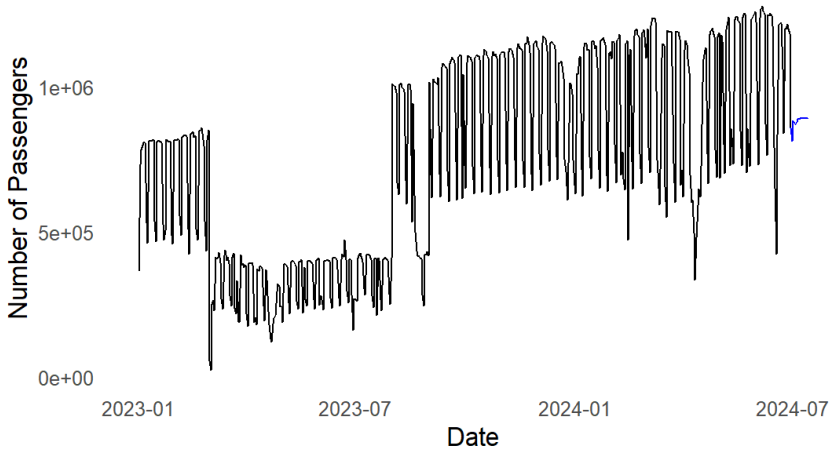


Figure 6. Plot actual data and 14-day forecast

CONCLUSION

The ARDL model produced is with optimum lag $p = 1$ and $q = 3$. This means that the number of Transjakarta passengers in the previous period, the previous two periods, and the previous three periods as well as route adjustments in the current period and the previous one have a significant impact on the number of Transjakarta passengers in the current period. The forecast for the first 14 days of July 2024 shows dynamic results, with a sharp increase in the initial forecast period followed by gradual stabilization.

SUGGESTION

This study provides several suggestions for consideration in future development. The model can be improved by adding other variables, such as traffic density levels, fare changes, or people's preferences for transportation modes. Additionally, conducting a study with a longer time frame is important to capture overall trends in Transjakarta usage, especially if new policies are introduced in the future. The forecasting results can also assist PT. Transportasi Jakarta in operational planning, such as adjusting schedules or allocating buses more effectively in specific corridors. Since route adjustments significantly impact passenger numbers, regular evaluations of route policies are necessary to ensure services remain satisfactory. Lastly, real-time data from smart card systems or passenger applications could be utilized to produce faster and more accurate predictions. By taking these steps, this study is expected to provide greater benefits, both in supporting the development of better public transportation and improving service quality for the community.

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